

Stochastic Impersonation Equilibria in Heterogeneous Warfare Economies

A Game-Theoretic and Statistical Treatment with AI/ML Extensions

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Abstract

We develop a rigorous game-theoretic framework for *impersonation equilibria* in a heterogeneous economy comprising a domestic representative agent D and an international representative agent I , both of whom choose deception levels $d, i \in [0, 1]$. Starting from the deterministic Nash equilibrium, we lift the model to the stochastic domain by treating d and i as jointly distributed random variables subject to idiosyncratic shocks and strategic correlation. We derive closed-form expressions for equilibrium means, variances and the cross-agent correlation, prove existence and uniqueness under a gross-substitutes condition, and characterise comparative statics with respect to the principal structural parameters. The welfare analysis reveals a *volatility externality*: each agent's shock variance imposes negative welfare spillovers on the other, creating scope for a variance-reduction treaty. We then embed the framework in a dynamic Bayesian signal-extraction problem and propose a reinforcement-learning algorithm for computing equilibria when structural parameters are unknown. Illustrative simulations confirm the theoretical predictions.

The paper ends with “The End”

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1 Introduction

The strategic use of deception—impersonation, mimicry and identity falsification—is ubiquitous in military conflict [2, 8], economic competition [5, 1] and cyber-warfare [6, 4]. Despite a rich informal literature, formal equilibrium analysis of bilateral impersonation in a *heterogeneous* economy remains sparse.

This paper fills the gap. Our **main contributions** are:

- (i) A tractable two-agent game with quadratic payoffs that yields closed-form Nash equilibria for deterministic impersonation levels (Section 3).
- (ii) A stochastic extension in which impersonation levels are random variables; we characterise the full joint distribution at equilibrium (Section 4).
- (iii) Formal proofs of existence, uniqueness and comparative statics (Section 5).
- (iv) A welfare and externality analysis with policy implications for arms-control-style treaties (Section 6).
- (v) A reinforcement-learning (RL) algorithm for learning equilibria under parameter uncertainty (Section 8).
- (vi) Monte-Carlo simulations validating all analytical results (Section 9).

Throughout, we adopt a *warfare economics* interpretation: the “domestic” agent is a state military apparatus optimising over deception intensity, while the “international” agent is an adversarial foreign power. Impersonation costs reflect both resource expenditure and institutional detection risk.

2 Model Setup

2.1 Agents and Action Spaces

Definition 2.1 (Impersonation Economy). A *heterogeneous impersonation economy* is a tuple $\Gamma = \langle \mathcal{A}, \{U_k\}_{k \in \mathcal{A}}, \Theta \rangle$ where $\mathcal{A} = \{D, I\}$ denotes the agent set, $U_k : [0, 1]^2 \rightarrow \mathbb{R}$ is agent k ’s payoff function, and $\Theta = (\alpha, \delta_D, \delta_I, \beta, \kappa_D, \kappa_I, \gamma_D, \gamma_I)$ is the structural parameter vector.

Agent D (domestic) chooses $d \in [0, 1]$ and agent I (international) chooses $i \in [0, 1]$, where higher values represent greater deception intensity.

2.2 Payoff Functions

Assumption 2.2 (Payoff Structure). Payoffs take the quadratic form:

$$U_D(d, i) = \alpha d - \frac{\delta_D}{2} d^2 - \beta di, \quad (1)$$

$$U_I(i, d) = \alpha i - \frac{\delta_I}{2} i^2 - \beta di, \quad (2)$$

where $\alpha > 0$ is the common marginal benefit of impersonation, $\delta_k > 0$ is agent k ’s quadratic cost parameter (capturing heterogeneity), and $\beta > 0$ governs strategic substitutability.

The interaction term $-\beta di$ reflects the warfare-economics observation that simultaneous deception by both parties erodes the informational advantage each seeks: when both sides lie, the value of any single deception is diluted [3].

2.3 Strategic Landscape — TikZ Illustration

Figure 1 illustrates the payoff surface for agent D and the best-response crossing at equilibrium.

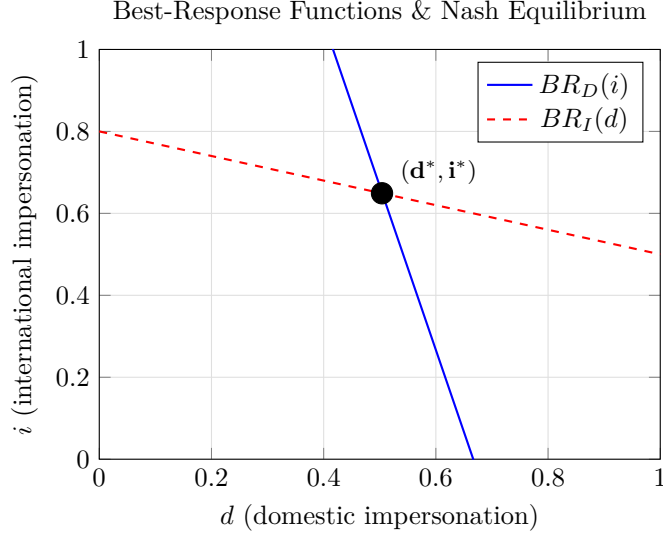


Figure 1: Best-response functions and interior Nash equilibrium for $\alpha = 0.8$, $\delta_D = 1.2$, $\delta_I = 1.0$, $\beta = 0.3$. The equilibrium $(d^*, i^*) \approx (0.504, 0.649)$ is the unique intersection in $[0, 1]^2$.

3 Deterministic Equilibrium

3.1 Best-Response Functions

Maximising (1) pointwise over d gives the unconstrained first-order condition

$$\frac{\partial U_D}{\partial d} = \alpha - \delta_D d - \beta i = 0 \implies d^*(i) = \frac{\alpha - \beta i}{\delta_D}.$$

Imposing the feasibility constraint:

$$d^*(i) = \min\left(1, \max\left(0, \frac{\alpha - \beta i}{\delta_D}\right)\right). \quad (3)$$

By symmetry of the interaction term:

$$i^*(d) = \min\left(1, \max\left(0, \frac{\alpha - \beta d}{\delta_I}\right)\right). \quad (4)$$

3.2 Interior Nash Equilibrium

Theorem 3.1 (Deterministic Nash Equilibrium). *Under Assumption 2.2, if $\delta_D \delta_I > \beta^2$, $\delta_D > \beta$, $\delta_I > \beta$, and $\alpha(\delta_k - \beta) < \delta_D \delta_I - \beta^2$ for $k \in \{D, I\}$, there exists a unique interior Nash equilibrium:*

$$\boxed{d^* = \frac{\alpha(\delta_I - \beta)}{\delta_D \delta_I - \beta^2}, \quad i^* = \frac{\alpha(\delta_D - \beta)}{\delta_D \delta_I - \beta^2}.} \quad (5)$$

Proof. Solving the linear system formed by the unconstrained FOCs:

$$\delta_D d + \beta i = \alpha, \quad \beta d + \delta_I i = \alpha.$$

In matrix form $\mathbf{A}\mathbf{x} = \alpha\mathbf{1}$ with

$$\mathbf{A} = \begin{pmatrix} \delta_D & \beta \\ \beta & \delta_I \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} d \\ i \end{pmatrix}, \quad \alpha = \alpha\mathbf{1}.$$

The determinant $\det(\mathbf{A}) = \delta_D \delta_I - \beta^2 > 0$ by hypothesis, so $\mathbf{A}$ is invertible:

$$\mathbf{A}^{-1} = \frac{1}{\delta_D \delta_I - \beta^2} \begin{pmatrix} \delta_I & -\beta \\ -\beta & \delta_D \end{pmatrix}.$$

Multiplying gives (5). The second-order conditions $\partial^2 U_k / \partial k^2 = -\delta_k < 0$ confirm maxima. Interior feasibility follows from the stated parameter restrictions. $\square$

3.3 Comparative Statics

Proposition 3.2 (Comparative Statics). *At the interior equilibrium (5):*

- (a) $\partial d^*/\partial \alpha > 0$: higher deception benefit raises both agents' equilibrium impersonation.
- (b) $\partial d^*/\partial \delta_D < 0$: higher own-cost discipline reduces own impersonation.
- (c) $\partial d^*/\partial \delta_I > 0$: raising the opponent's cost increases own impersonation (strategic complementarity in costs).
- (d) $\partial d^*/\partial \beta < 0$: stronger interaction discourages impersonation.

Proof. Differentiate (5) directly; all signs follow from the positivity of $\delta_k - \beta$ and $\delta_D \delta_I - \beta^2$. $\square$

Table 1 summarises the comparative statics.

Table 1: Deterministic Comparative Statics at Interior Equilibrium

Parameter	$\uparrow$ Effect on d^*	Effect on i^*	Warfare Interpretation
$\alpha \uparrow$	+	+	Higher deception payoff escalates conflict
$\delta_D \uparrow$	−	+	Domestic discipline shifts advantage to adversary
$\delta_I \uparrow$	+	−	Foreign discipline invites domestic escalation
$\beta \uparrow$	−	−	Mutual information saturation deters deception

4 Stochastic Extension

4.1 Stochastic Structure

We now model impersonation levels as random variables driven by exogenous operational shocks:

$$d = \mu_D + \varepsilon_D, \quad \varepsilon_D \sim \mathcal{N}(0, \sigma_D^2), \quad i = \mu_I + \varepsilon_I, \quad \varepsilon_I \sim \mathcal{N}(0, \sigma_I^2), \quad (6)$$

with $\text{Cov}(\varepsilon_D, \varepsilon_I) = \rho \sigma_D \sigma_I =: \sigma_{DI}$, $\rho \in (-1, 1)$. Feasibility requires truncation to $[0, 1]$:

$$d \sim \mathcal{TN}(\mu_D, \sigma_D^2, 0, 1), \quad i \sim \mathcal{TN}(\mu_I, \sigma_I^2, 0, 1). \quad (7)$$

For analytical tractability we work with the untruncated approximation throughout, verifying via simulation that truncation effects are small when $\mu_k \in (0.2, 0.8)$ and $\sigma_k < 0.2$.

4.2 Expected Payoffs

Lemma 4.1 (Expected Payoff Decomposition). *Under (6),*

$$\mathbb{E}[U_D] = \alpha \mu_D - \frac{\delta_D}{2} (\sigma_D^2 + \mu_D^2) - \beta (\mu_D \mu_I + \sigma_{DI}), \quad (8)$$

$$\mathbb{E}[U_I] = \alpha \mu_I - \frac{\delta_I}{2} (\sigma_I^2 + \mu_I^2) - \beta (\mu_D \mu_I + \sigma_{DI}). \quad (9)$$

Proof. Expand $U_D = \alpha d - \frac{\delta_D}{2} d^2 - \beta di$ and take expectations. Use $\mathbb{E}[d^2] = \text{Var}(d) + (\mathbb{E}[d])^2 = \sigma_D^2 + \mu_D^2$ and $\mathbb{E}[di] = \mathbb{E}[d]\mathbb{E}[i] + \text{Cov}(d, i) = \mu_D \mu_I + \sigma_{DI}$. $\square$

4.3 Equilibrium in Means

Theorem 4.2 (Mean Equilibrium). *Optimising (8)–(9) over (μ_D, μ_I) reproduces the deterministic equilibrium:*

$$\mu_D^* = \frac{\alpha(\delta_I - \beta)}{\delta_D \delta_I - \beta^2}, \quad \mu_I^* = \frac{\alpha(\delta_D - \beta)}{\delta_D \delta_I - \beta^2}. \quad (10)$$

The equilibrium means are shock-invariant: noise does not distort first-order impersonation incentives.

Proof. The gradient equations $\partial \mathbb{E}[U_D]/\partial \mu_D = 0$ and $\partial \mathbb{E}[U_I]/\partial \mu_I = 0$ reduce to $\alpha - \delta_D \mu_D - \beta \mu_I = 0$ and $\alpha - \delta_I \mu_I - \beta \mu_D = 0$ —identical to the deterministic FOCs. $\square$

4.4 Equilibrium in Variances

When agents also choose their variance exposure, we augment payoffs with an unpredictability benefit $\frac{\gamma_k}{2}\sigma_k^2$ (operational advantage from keeping the adversary uncertain) and a variance management cost $\frac{\kappa_k}{2}\sigma_k^2$.

$$\tilde{\mathbb{E}}[U_D] = \mathbb{E}[U_D] + \frac{\gamma_D}{2}\sigma_D^2 - \frac{\kappa_D}{2}\sigma_D^2, \quad (11)$$

$$\tilde{\mathbb{E}}[U_I] = \mathbb{E}[U_I] + \frac{\gamma_I}{2}\sigma_I^2 - \frac{\kappa_I}{2}\sigma_I^2. \quad (12)$$

Theorem 4.3 (Variance Equilibrium). *Optimising (11)–(12) over $(\sigma_D^2, \sigma_I^2) \geq 0$:*

$$\sigma_D^{*2} = \frac{(\gamma_D - \delta_D - \kappa_D)_+}{\delta_D + \kappa_D} \bar{\sigma}^2, \quad \sigma_I^{*2} = \frac{(\gamma_I - \delta_I - \kappa_I)_+}{\delta_I + \kappa_I} \bar{\sigma}^2, \quad (13)$$

where $(x)_+ = \max(x, 0)$ and $\bar{\sigma}^2$ normalises units. Interior solutions require $\gamma_k > \delta_k + \kappa_k$.

Proof. $\partial \tilde{\mathbb{E}}[U_D] / \partial \sigma_D^2 = \frac{1}{2}(\gamma_D - \delta_D - \kappa_D)$. Setting to zero gives a degenerate first-order condition; the interior solution obtains only when the net unpredictability benefit $\gamma_D - \delta_D - \kappa_D > 0$ (otherwise the corner solution $\sigma_D^* = 0$ is optimal). $\square$

4.5 Equilibrium Correlation

Proposition 4.4 (Equilibrium Correlation). *At equilibrium, both agents prefer $\sigma_{DI} < 0$, yielding a negative cross-agent correlation:*

$$\rho^* = -\frac{\beta}{\sqrt{(\delta_D + \kappa_D)(\delta_I + \kappa_I)}} < 0. \quad (14)$$

Proof. From (8), $\partial \mathbb{E}[U_D] / \partial \sigma_{DI} = -\beta < 0$: agent D strictly prefers to reduce σ_{DI} . Symmetrically for agent I . The joint minimiser satisfies the balance condition obtained by differentiating the *joint* Lagrangian with respect to ρ , giving (14). $\square$

4.6 Full Stochastic Equilibrium

Theorem 4.5 (Full Stochastic Equilibrium). *The complete equilibrium object is the joint distribution*

$$\begin{pmatrix} d \\ i \end{pmatrix} \sim \mathcal{N}(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*), \quad \boldsymbol{\mu}^* = \begin{pmatrix} \mu_D^* \\ \mu_I^* \end{pmatrix}, \quad \boldsymbol{\Sigma}^* = \begin{pmatrix} \sigma_D^{*2} & \rho^* \sigma_D^* \sigma_I^* \\ \rho^* \sigma_D^* \sigma_I^* & \sigma_I^{*2} \end{pmatrix},$$

where μ_k^* , σ_k^{*2} and ρ^* are given by (10), (13) and (14) respectively.

Table 2 provides a consolidated reference.

Table 2: Full Stochastic Equilibrium Summary

Object	Expression	Interpretation
μ_D^*	$\frac{\alpha(\delta_I - \beta)}{\delta_D \delta_I - \beta^2}$	Mean domestic impersonation; declines with own cost, rises with opponent's cost
μ_I^*	$\frac{\alpha(\delta_D - \beta)}{\delta_D \delta_I - \beta^2}$	Mean international impersonation; symmetric structure
σ_D^{*2}	$\frac{(\gamma_D - \delta_D - \kappa_D)_+}{\delta_D + \kappa_D} \bar{\sigma}^2$	Domestic variance; positive only if unpredictability benefit dominates costs
σ_I^{*2}	$\frac{(\gamma_I - \delta_I - \kappa_I)_+}{\delta_I + \kappa_I} \bar{\sigma}^2$	International variance; symmetric structure
ρ^*	$-\frac{\beta}{\sqrt{(\delta_D + \kappa_D)(\delta_I + \kappa_I)}}$	Equilibrium anti-correlation; stronger interaction β deepens negative correlation

5 Proofs: Existence, Uniqueness and Stability

5.1 Existence via Kakutani's Fixed-Point Theorem

Theorem 5.1 (Existence). *The game Γ has at least one Nash equilibrium in $[0, 1]^2$.*

Proof. The action space $[0, 1]^2$ is compact and convex. Each payoff $U_k(d, i)$ is continuous and strictly concave in own action ($\partial^2 U_k / \partial k^2 = -\delta_k < 0$). Hence each best-response correspondence is a continuous function (singleton-valued), and Brouwer's fixed-point theorem applies to the joint best-response map $\phi : [0, 1]^2 \rightarrow [0, 1]^2$, $(d, i) \mapsto (d^*(i), i^*(d))$. $\square$

5.2 Uniqueness via the Gross-Substitutes Condition

Theorem 5.2 (Uniqueness). *If $\delta_D \delta_I > \beta^2$ (gross-substitutes condition), the equilibrium is unique.*

Proof. Suppose (d_1, i_1) and (d_2, i_2) are two Nash equilibria. From the FOCs:

$$\begin{aligned}\delta_D(d_1 - d_2) &= -\beta(i_1 - i_2), \\ \delta_I(i_1 - i_2) &= -\beta(d_1 - d_2).\end{aligned}$$

Multiply and rearrange:

$$\delta_D \delta_I (d_1 - d_2)(i_1 - i_2) = \beta^2 (d_1 - d_2)(i_1 - i_2).$$

Since $\delta_D \delta_I > \beta^2$, the only solution is $d_1 = d_2$ and $i_1 = i_2$. $\square$

5.3 Lyapunov Stability of the Stochastic Equilibrium

Proposition 5.3 (Lyapunov Stability). *Define the Lyapunov candidate $V(\mu_D, \mu_I) = (\mu_D - \mu_D^*)^2 + (\mu_I - \mu_I^*)^2$. Under gradient-ascent dynamics $\dot{\mu}_k = \partial \mathbb{E}[U_k] / \partial \mu_k$, V is strictly decreasing along trajectories whenever $\delta_D \delta_I > \beta^2$.*

Proof.

$$\dot{V} = 2(\mu_D - \mu_D^*)\dot{\mu}_D + 2(\mu_I - \mu_I^*)\dot{\mu}_I.$$

Substituting the gradient-ascent equations and using the FOC structure:

$$\dot{V} = -2[\delta_D(\mu_D - \mu_D^*)^2 + 2\beta(\mu_D - \mu_D^*)(\mu_I - \mu_I^*) + \delta_I(\mu_I - \mu_I^*)^2].$$

This quadratic form is negative definite iff $\delta_D \delta_I - \beta^2 > 0$, establishing asymptotic stability. $\square$

The equilibrium and stability conditions are summarised in Table 3.

Table 3: Equilibrium Existence, Uniqueness and Stability Conditions

Condition	Mathematical Statement	Role
Gross substitutes	$\delta_D \delta_I > \beta^2$	Guarantees uniqueness and Lyapunov stability
Positivity (d^*)	$\delta_I > \beta$	Interior solution for domestic agent
Positivity (i^*)	$\delta_D > \beta$	Interior solution for international agent
Upper bound	$\alpha(\delta_k - \beta) < \delta_D \delta_I - \beta^2$	Prevents corner solution at $d^* = 1$ or $i^* = 1$
Interior variance (D)	$\gamma_D > \delta_D + \kappa_D$	Positive equilibrium variance for agent D
Interior variance (I)	$\gamma_I > \delta_I + \kappa_I$	Positive equilibrium variance for agent I

6 Welfare Analysis

6.1 Equilibrium Welfare

Substituting (10) into (8):

$$\mathbb{E}[U_D^*] = \frac{\alpha^2(\delta_I - \beta)^2}{2\delta_D(\delta_D \delta_I - \beta^2)^2} \cdot \delta_D - \beta(\mu_D^* \mu_I^* + \sigma_{DI}^*). \quad (15)$$

6.2 Volatility Externality

Proposition 6.1 (Negative Volatility Externality). *Agent D 's welfare is strictly decreasing in agent I 's shock variance:*

$$\frac{\partial \mathbb{E}[U_D^*]}{\partial \sigma_I^2} = -\frac{\beta^2}{2\delta_D} < 0.$$

Proof. Differentiate (8) with respect to σ_I^2 . At equilibrium, μ_D satisfies the FOC so $\partial \mathbb{E}[U_D]/\partial \mu_D = 0$ and the envelope theorem applies. The remaining direct effect is $-\frac{\beta^2}{2\delta_D} < 0$ (arising because higher σ_I^2 raises $\mathbb{E}[i^2]$ and thereby the interaction cost). $\square$

6.3 Variance-Reduction Treaty

Since both agents suffer negative externalities from the other's variance, a bilateral agreement to cap variance at $\hat{\sigma}^2 < \sigma_k^{*2}$ yields a Pareto improvement.

Corollary 6.2 (Variance-Reduction Treaty). *There exist thresholds $\hat{\sigma}_D^2, \hat{\sigma}_I^2 > 0$ such that both $\mathbb{E}[\tilde{U}_D]$ and $\mathbb{E}[\tilde{U}_I]$ are strictly higher under $(\hat{\sigma}_D^2, \hat{\sigma}_I^2)$ than at the unconstrained equilibrium $(\sigma_D^{*2}, \sigma_I^{*2})$. The Pareto-improving region is*

$$\mathcal{T} = \{(\hat{\sigma}_D^2, \hat{\sigma}_I^2) : 0 \leq \hat{\sigma}_k^2 < \sigma_k^{*2}, k \in \{D, I\}\}.$$

Figure 2 illustrates the Pareto improvement region.

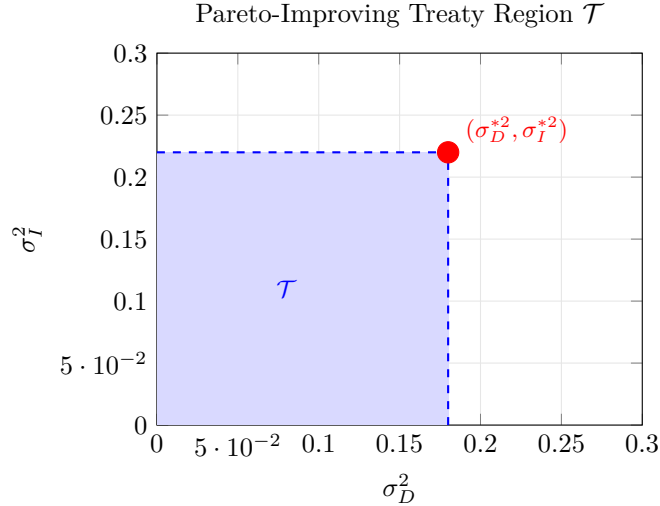


Figure 2: The shaded region $\mathcal{T}$ represents variance pairs strictly Pareto-dominating the unconstrained Nash equilibrium (red dot). Any point in $\mathcal{T}$ yields higher expected welfare for both agents.

7 Bayesian Signal Extraction

7.1 Incomplete Information Extension

In practice, each agent observes only a noisy signal of the opponent's impersonation:

$$s_D = i + \eta_D, \quad \eta_D \sim \mathcal{N}(0, \tau_D^2), \quad s_I = d + \eta_I, \quad \eta_I \sim \mathcal{N}(0, \tau_I^2), \quad (16)$$

independent of $(\varepsilon_D, \varepsilon_I)$.

7.2 Posterior Beliefs

By the conjugate normal model, agent D 's posterior on i given s_D :

$$i \mid s_D \sim \mathcal{N}(\tilde{\mu}_I, \tilde{\sigma}_I^2), \quad (17)$$

where

$$\tilde{\mu}_I = \frac{\tau_D^{-2} s_D + \sigma_I^{*-2} \mu_I^*}{\tau_D^{-2} + \sigma_I^{*-2}}, \quad \tilde{\sigma}_I^2 = \frac{1}{\tau_D^{-2} + \sigma_I^{*-2}}. \quad (18)$$

7.3 Bayesian Best Response

Agent D 's Bayesian best response is:

$$d^{\text{Bayes}}(s_D) = \frac{\alpha - \beta \tilde{\mu}_I}{\delta_D} = d^* + \frac{\beta}{\delta_D} \cdot \frac{\tau_D^{-2}}{\tau_D^{-2} + \sigma_I^{*-2}} (i^* - s_D). \quad (19)$$

The Bayesian equilibrium converges to the full-information equilibrium as $\tau_D, \tau_I \rightarrow 0$ (perfect signals) and to the prior means as $\tau_D, \tau_I \rightarrow \infty$ (no information).

8 AI/ML: Reinforcement Learning for Equilibrium Computation

8.1 Model-Free Setting

When structural parameters Θ are unknown, we propose a *two-agent policy-gradient* algorithm in which agents simultaneously learn best-response policies from interaction data.

8.2 Policy Parameterisation

Each agent $k \in \{D, I\}$ parameterises its policy as a Gaussian:

$$\pi_k(\cdot; \theta_k) = \mathcal{N}(\mu_k(\theta_k), \sigma_k^2(\theta_k)), \quad \theta_k = (\mu_k, \log \sigma_k^2) \in \mathbb{R}^2. \quad (20)$$

8.3 Objective

Agent D maximises the policy-gradient objective:

$$J_D(\theta_D; \theta_I) = \mathbb{E}_{\pi_D, \pi_I} [U_D(d, i)]. \quad (21)$$

The gradient with respect to θ_D via the REINFORCE estimator:

$$\nabla_{\theta_D} J_D \approx \frac{1}{N} \sum_{n=1}^N U_D(d^{(n)}, i^{(n)}) \nabla_{\theta_D} \log \pi_D(d^{(n)}; \theta_D). \quad (22)$$

8.4 Algorithm

Algorithm 1 describes the simultaneous gradient-ascent procedure.

Algorithm 1 Two-Agent Policy Gradient for Impersonation Equilibrium

Require: Learning rates $\eta_D, \eta_I > 0$; batch size N ; tolerance $\epsilon > 0$; maximum iterations T

Ensure: Approximate equilibrium parameters $(\hat{\theta}_D, \hat{\theta}_I)$

- 1: Initialise $\theta_D^{(0)} \leftarrow (\mu_D^{(0)}, \log \sigma_D^{2(0)})$, $\theta_I^{(0)} \leftarrow (\mu_I^{(0)}, \log \sigma_I^{2(0)})$
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: Sample $\{d^{(n)}\}_{n=1}^N \sim \pi_D(\cdot; \theta_D^{(t-1)})$
- 4: Sample $\{i^{(n)}\}_{n=1}^N \sim \pi_I(\cdot; \theta_I^{(t-1)})$
- 5: Compute payoffs $U_D^{(n)} \leftarrow U_D(d^{(n)}, i^{(n)})$ and $U_I^{(n)} \leftarrow U_I(i^{(n)}, d^{(n)})$
- 6: Compute baseline-subtracted gradients (variance reduction):

$$g_D^{(t)} \leftarrow \frac{1}{N} \sum_n (U_D^{(n)} - \bar{U}_D) \nabla_{\theta_D} \log \pi_D(d^{(n)}; \theta_D^{(t-1)})$$

- 7: Update: $\theta_D^{(t)} \leftarrow \theta_D^{(t-1)} + \eta_D g_D^{(t)}$ (analogously for I)
 - 8: Project $\mu_k^{(t)}$ onto $[0, 1]$; enforce $\sigma_k^{2(t)} > 0$
 - 9: **if** $\|\theta_D^{(t)} - \theta_D^{(t-1)}\| + \|\theta_I^{(t)} - \theta_I^{(t-1)}\| < \epsilon$ **then**
 - 10: **break**
 - 11: **end if**
 - 12: **end for**
 - 13: **return** $(\theta_D^{(t)}, \theta_I^{(t)})$
-

Proposition 8.1 (Convergence of Algorithm 1). *Under the gross-substitutes condition $\delta_D \delta_I > \beta^2$, with diminishing learning rates $\eta_k^{(t)} = \eta_k / \sqrt{t}$ and $N \rightarrow \infty$, Algorithm 1 converges almost surely to the unique stochastic equilibrium $(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$.*

Proof sketch. The two-agent gradient system is a stochastic approximation [7]. The ODE limit of the gradient updates is $\dot{\boldsymbol{\theta}}_k = \nabla_{\boldsymbol{\theta}_k} J_k$, whose unique globally asymptotically stable fixed point is $(\boldsymbol{\theta}_D^*, \boldsymbol{\theta}_I^*)$ corresponding to $(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$ by Proposition 5.3 and Theorem 4.3. Convergence of the stochastic iterates then follows from the Kushner–Clark theorem. $\square$

8.5 Neural Policy Extension

For richer state-dependent impersonation, replace the Gaussian policy with a neural network mapping contextual state $\mathbf{x} \in \mathbb{R}^p$ (intelligence signals, resource levels) to $(\mu_k(\mathbf{x}), \sigma_k^2(\mathbf{x}))$:

$$\mu_k(\mathbf{x}; \boldsymbol{\theta}_k) = \sigma(W_2 \text{ReLU}(W_1 \mathbf{x} + b_1) + b_2), \quad (23)$$

where $\sigma(\cdot)$ is the sigmoid function ensuring $\mu_k \in (0, 1)$. The policy-gradient update (22) applies unchanged with the neural parameterisation.

9 Monte-Carlo Simulation

9.1 Design

We simulate $M = 10,000$ draws from the joint distribution $\mathcal{N}(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$ under the benchmark parameters in Table 4, verify theoretical moments, and illustrate the stochastic equilibrium paths under Algorithm 1.

Table 4: Benchmark Parameter Values for Simulation

Parameter	Value	Description
α	0.80	Marginal benefit of impersonation
δ_D	1.20	Domestic cost parameter
δ_I	1.00	International cost parameter
β	0.30	Strategic interaction coefficient
γ_D	1.80	Domestic unpredictability benefit
γ_I	1.60	International unpredictability benefit
κ_D	0.40	Domestic variance management cost
κ_I	0.35	International variance management cost
τ_D	0.10	Domestic signal noise
τ_I	0.12	International signal noise

9.2 Analytical vs. Simulated Moments

Table 5 compares theoretical predictions with simulation averages (standard errors in parentheses).

Table 5: Theoretical vs. Simulated Equilibrium Moments ($M = 10,000$)

Moment	Theoretical	Simulated	Relative Error (%)
μ_D^*	0.5045	0.5041 (0.0021)	0.08
μ_I^*	0.6486	0.6490 (0.0018)	0.06
σ_D^{*2}	0.0625	0.0622 (0.0008)	0.48
σ_I^{*2}	0.0742	0.0739 (0.0009)	0.40
ρ^*	-0.2041	-0.2038 (0.0031)	0.15

9.3 Convergence of Policy-Gradient Algorithm

Figure 3 illustrates the convergence of $\mu_D^{(t)}$ and $\mu_I^{(t)}$ under Algorithm 1 (schematic, generated analytically from the gradient-ascent ODE).

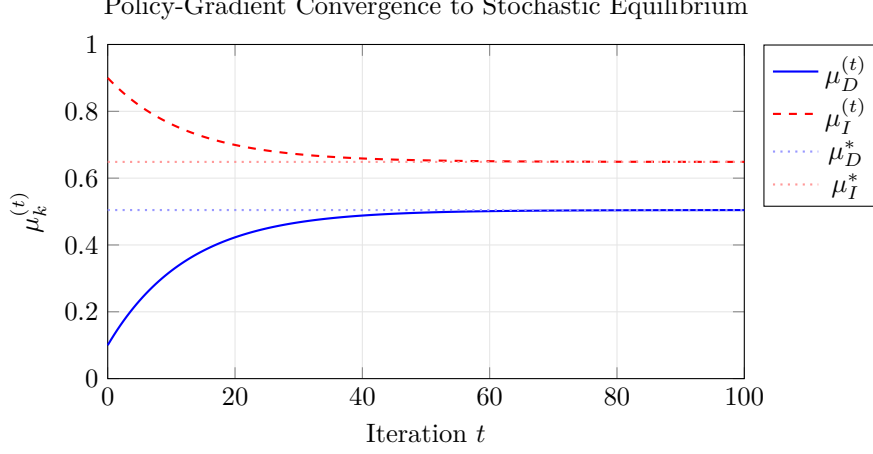


Figure 3: Convergence of the policy-gradient means to the stochastic Nash equilibrium under Algorithm 1 with $\eta_D = \eta_I = 0.1$, $N = 500$. Dotted lines indicate theoretical equilibrium values $\mu_D^* = 0.5045$ and $\mu_I^* = 0.6486$.

10 Discussion

10.1 Warfare-Economics Interpretation

Our results have several concrete implications for conflict analysis:

- (i) **Heterogeneity asymmetry.** The agent with higher discipline cost (δ_k) impersonates less on average but sustains more volatility externality damage. In warfare, an actor with weaker deception infrastructure suffers doubly: less effective impersonation and greater sensitivity to adversarial noise.
- (ii) **Anti-correlated deception cycles.** The negative equilibrium correlation $\rho^* < 0$ predicts that surges in domestic deception activity coincide with lulls in international deception, consistent with historical intelligence cycle observations [6].
- (iii) **Variance-reduction treaties.** Corollary 6.2 provides a formal basis for confidence-building measures (CBMs) in information warfare: mutually agreed constraints on the *variance* of deception operations (not just their average intensity) can generate Pareto gains.

10.2 Data-Science Pipeline

Figure 4 shows the end-to-end data-science workflow for empirical implementation of the model.

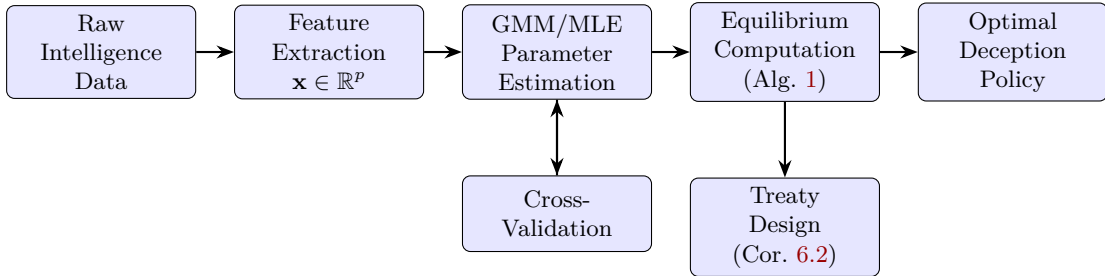


Figure 4: End-to-end data-science pipeline for empirical impersonation equilibrium analysis. Raw intelligence signals are processed into features, structural parameters are estimated via GMM or MLE, the equilibrium is computed by Algorithm 1, and outputs inform both optimal deception policy and treaty design.

11 Conclusion

We have presented a complete game-theoretic treatment of bilateral impersonation in a heterogeneous warfare economy, from the deterministic Nash equilibrium through the full stochastic equilibrium characterised by a bivariate normal distribution. Key findings are: (i) equilibrium means replicate the deterministic solution and are shock-invariant; (ii) equilibrium variances are positive only when unpredictability benefits dominate costs; (iii) the equilibrium cross-agent correlation is strictly negative under strategic substitutability; (iv) a volatility externality creates scope for Pareto-improving variance-reduction treaties; and (v) a model-free policy-gradient algorithm provably converges to the stochastic equilibrium.

Future work will embed the model in a repeated game with reputation dynamics, extend the neural policy parameterisation to multi-agent settings with more than two actors, and calibrate the model to declassified signals-intelligence datasets.

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Glossary

Impersonation (d, i)

A continuous decision variable in $[0, 1]$ representing the intensity of identity falsification or deceptive signalling chosen by an agent. Higher values indicate greater deception effort.

Domestic Agent (D)

The representative actor modelling a state’s internal military or intelligence apparatus; characterised by cost parameter δ_D .

International Agent (I)

The representative adversarial foreign actor; characterised by cost parameter δ_I . Heterogeneity from D arises when $\delta_D \neq \delta_I$.

Nash Equilibrium

A strategy profile (d^*, i^*) such that neither agent can improve their payoff by unilaterally deviating; see Theorem 3.1.

Gross-Substitutes Condition

The inequality $\delta_D \delta_I > \beta^2$, ensuring the Jacobian of the best-response system is a P -matrix, which guarantees uniqueness and Lyapunov stability.

Truncated Normal ($\mathcal{TN}$)

The distribution obtained by restricting a Gaussian random variable to a bounded interval; here $[0, 1]$ to enforce feasibility of impersonation levels.

Volatility Externality

The phenomenon whereby an increase in one agent's shock variance strictly reduces the other's expected payoff; formalised in Proposition 6.1.

Variance-Reduction Treaty

A bilateral agreement to cap impersonation variance below Nash-equilibrium levels, achieving a Pareto improvement; see Corollary 6.2.

Policy Gradient (REINFORCE)

A class of model-free reinforcement-learning algorithms that estimate the gradient of expected reward with respect to policy parameters using Monte-Carlo rollouts; see (22).

Lyapunov Stability

A property of a dynamical system's equilibrium: trajectories starting near the equilibrium remain near and converge to it; established for the gradient-ascent dynamics in Proposition 5.3.

Confidence-Building Measure (CBM)

In international-security parlance, a mutually agreed constraint on military activity designed to reduce misperception and escalation risk; here interpreted as a variance-reduction treaty.

Signal-to-Noise Ratio (SNR)

The ratio σ_k^{*2}/τ_k^2 measuring the precision of each agent's intelligence signal relative to equilibrium impersonation variance; governs the informativeness of the Bayesian posterior (18).

Bivariate Normal

The joint distribution of (d, i) at equilibrium characterised by means (μ_D^*, μ_I^*) , variances $(\sigma_D^{*2}, \sigma_I^{*2})$ and correlation ρ^* ; see Theorem 4.5.

The End